

Supplementary material for: Information loss of fixed linear health indicators under competing degradation mechanisms and variable operating conditions

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Abstract

Section S1 gives the protocol behind every row of Table 4 of the main text and behind its case study on the Severson cells, together with the parameters of the code that produced them. Section S2 derives the loss over a complete record and characterises the best fixed indicator as a barycentre; Section S3 reports the numerical verification of the identities and examines the non-generic systems to which the counting condition does not apply; and Section S4 contains the conditions for zero loss and further results on the synthetic system that the main text summarises. Equation, section, figure and table numbers without the prefix S refer to the main text.

S1 Protocols for the public-fleet tests

Every number in Table 4 is recomputed by `verify_numbers.py` in the repository named in the main text, and the parameters below are those passed by that script; where a test was also run at other settings to check stability, both are given. Every test below that uses an informative direction estimates it as a time derivative of the monitoring channels along normalised life, and therefore relies on the time-shift population of Section 2, under which that derivative is the sensitivity of Eq. (3).

S1.1 Rotation test on the Severson cells

The channels are discharge capacity, internal resistance and mean cell temperature. A cell is used if its life is recorded, it has at least 400 cycles, and all three channels are finite throughout; 129 cells qualify. Life is normalised per cell to $\tau \in [0, 1]$ by cycle index. Each channel is differentiated with a Savitzky–Golay filter of polynomial order 3 and window $\min(301, 2\lfloor n/4 \rfloor + 1)$ cycles for a cell of n cycles, and the derivative is divided by the standard deviation of that channel’s residual about the same filter, which whitens the channel by its own measured noise. The whitened derivative is interpolated onto 40 points of $\tau \in [0.15, 0.90]$. One sign per channel, the median sign over every cell and grid point, orients all cells consistently; each cell’s vector is then normalised at every grid point, and the renormalised fleet mean is the empirical informative

direction. Its arc is the sum of the angles between consecutive grid points, and its end-to-end angle is the angle between the first and the last. Positivity is the fraction of grid points at which every component of the fleet mean is positive; it equals one at the window used.

S1.2 Remaining-life test on the Severson cells

The test uses the cells, smoothing, whitening and orientation of the rotation test and is run leave-one-out: for each test cell, the reference path, the noise levels and the fixed indicator are computed from the other 128 cells. Each channel is referred to its own baseline, the median of its first 50 cycles. The reference path is the median over cells of the smoothed, baseline-referred channels against normalised life, tabulated at 1001 points, and the noise level of each channel is the median over cells of its residual standard deviation. The fixed indicator is the renormalised mean over $\tau \in [0.15, 0.90]$ of the fleet-mean informative direction. At an elapsed cycle count n_0 , the integer nearest to $\tau_0(n - 1)$ for a cell of n cycles with $\tau_0 \in \{0.5, 0.7, 0.9\}$, the estimator observes the cycles from $0.7n_0$ to n_0 and returns the life L , searched over the integers from $n_0 + 1$ to 4000, for which the reference path evaluated at each observed cycle divided by $L - 1$ is closest to the observations in whitened least squares; the error is $(L - n)/n$. The estimator is run on the three whitened channels, on the fixed indicator, on capacity alone, on temperature alone and on a fusion indicator, and each single-signal estimator is compared with the three-channel one through the ratio of their mean squared errors over cells. The fusion weights are $C^{-1}\bar{\Delta}$, normalised to unit length, where $\bar{\Delta}$ and C are the mean and the between-cell covariance, over the training cells, of the whitened, oriented, baseline-referred smoothed channels at failure; they make the indicator's value at failure as consistent between cells as possible relative to its mean change over life, the criterion of data-level fusion [1]. Observing the last 20% or 50% of the elapsed cycles instead, or taking the baseline over the first 20 or 100 cycles, keeps that ratio, pooled over the three ages, between 0.95 and 0.97 for the fixed indicator and between 1.03 and 1.06 for capacity alone.

S1.3 Replacement policy on the Severson cells

Each cell is inspected every 25 cycles from cycle 100 until its end of life. At each inspection the estimator of the remaining-life test, with the life searched from $n_0 + 1$ to $n_0 + 1500$, predicts the remaining life, and the cell is replaced at the first inspection at which the prediction is at most a threshold of n_{thr} cycles; a cell not replaced before its end of life fails. With r the ratio of the corrective to the preventive cost, the long-run cost per cycle is $(r n_f + n_p) / \sum_k T_k$, where n_f and n_p count failures and preventive replacements and T_k is the service time of cell k . The threshold is chosen on a grid from 0 to 1000 cycles in steps of 5 to minimise this cost, separately for each signal. At $r = 10$ the optimal threshold is 130 cycles for all three channels, the fixed indicator and capacity alone, and 120 cycles for the fusion indicator. At the smallest threshold that keeps failures at or below 5%, which is 90 cycles for every signal, the mean unused life is 92 cycles for all three channels, the fixed indicator and capacity alone, and 112 cycles for the fusion indicator.

S1.4 Measured fleets that do not permit a test of actuation

A test of actuation requires the operating point to vary within a unit while all other conditions are held constant. In XJTU-SY each of three operating conditions is applied to its own five bearings, and in PRONOSTIA each of three conditions to its own bearings, so a difference

between conditions cannot be distinguished from a difference between groups of units. The NASA battery fleet has four cells. These fleets are therefore listed with the reason and without a statistic.

S1.5 Regime contrast on C-MAPSS

The training files of FD004 and FD002 are used. Life is normalised per engine by its final cycle. Operating regimes are labelled by standardising the three operating settings over the file and rounding to one decimal; each file has 6 regimes. A sensor is retained if its standard deviation within an (engine, regime) group has a positive median, and each retained sensor is standardised within each regime, so that the level of a regime does not enter the direction.

Within one engine, the cycles of each regime are resampled to a common count m , the smallest over that engine's regimes, and an engine enters only if $m \geq 2(w+3)$. Each resampled series is split into its even and its odd cycles, two half-samples of the same regime. On each half the direction is estimated as for the Severson cells, with polynomial order 2 and window w , and interpolated onto 10 points of $\tau \in [0.30, 0.85]$. The angle between two directions is the mean over grid points of the pointwise angle. W is the mean, over the engine's regimes, of the angle between the two halves of the same regime, and B the mean, over pairs of different regimes, of the angle between a half of one and a half of the other. $B - W$ is positive when regimes differ by more than half-samples of one regime differ from each other, as predicted by an operating point that moves the observed direction. The contrast does not, however, indicate how the direction is moved. C-MAPSS imposes its flow and efficiency losses as functions of time and lets the regime act only on the thermodynamic response of the sensors [2], so a positive contrast there reflects the observation map, not the stress factor of Eq. (1), and cannot test actuation. The table reports the mean of $B - W$ over engines, and the number of engines for which it is positive, at $w = 7$; at $w = 9$ and $w = 11$ every value keeps its sign and exceeds the largest control bias.

The negative control applies the same procedure to surrogate regimes. Within one regime of one engine with at least $4(w+3)$ cycles, the four interleaved subsequences of every fourth cycle serve as four surrogate regimes; W compares the first with the second and B the first with the fourth. No contrast exists by construction. Over all three windows and both files the mean of $B - W$ stays within 0.5° of zero, and a paired t test of B against W does not reject equality at any window, the smallest p being 0.35.

S1.6 Joint test of separability and channel pairing on N-CMAPSS

The development sets of DS02, units 2, 5, 10, 16, 18 and 20, and of DS01, units 1 to 6, are used. The operating point is held fixed by clustering flight points, by k -means with 6 clusters, on standardised altitude, Mach number and throttle-resolver angle; life is divided into 2 stages of equal width in cycles. A (stage, cluster) cell enters only with at least 400 points.

Within a cell each sensor is regressed on standardised remaining life, and the slope is divided by the standard deviation of that sensor's residual, so that the vector of slopes is the informative direction at a fixed operating point rather than a residual after standardisation. The vector is normalised, and the test compares the logarithms of the absolute values of its components, centred on their mean. The logarithmic transformation is essential: under the separable form of Section S4.1, interpreted through the pairing of channels with mechanisms in Eq. (4), each component is the product of an operating-point factor and a state factor, so in logarithms the two act additively, and the absence of interaction is what the two assumptions jointly predict. A sensor map that mixes mechanisms with weights depending on the operating point violates

Table S1: Parameters of each test as run.

test	parameters
Severson rotation	129 cells with at least 400 cycles; Savitzky–Golay order 3, window 301 cycles or $2\lfloor n/4 \rfloor + 1$ if smaller; 40 points on $\tau \in [0.15, 0.90]$
Severson remaining life	the same cells, leave-one-out; baseline the median of the first 50 cycles; reference path on 1001 points; the last 30% of the elapsed cycles at $\tau_0 = 0.5, 0.7, 0.9$; life searched up to 4000 cycles; fusion weights from the channels at failure
Severson replacement policy	inspections every 25 cycles from cycle 100; life searched up to 1500 cycles beyond the inspection; threshold grid 0 to 1000 cycles in steps of 5; cost ratios 5, 10, 20 and 50
C-MAPSS regime contrast	FD004 and FD002 training sets; 6 regimes each; order 2, window $w = 7$, stability at $w = 9, 11$; 10 points on $\tau \in [0.30, 0.85]$; engine enters if $m \geq 2(w + 3)$; control needs $4(w + 3)$ cycles in one regime
N-CMAPSS separability with the pairing	DS02 units 2, 5, 10, 16, 18, 20 and DS01 units 1–6; 6 clusters, 2 stages, at least 400 points per cell; sweep over seven (cluster, stage) settings with at least 300 points

this additivity just as a non-separable stress factor does. In N-CMAPSS the operating history enters the damage through the onset of abnormal degradation [3], and the health parameters supplied with the files, which are the damage imposed by the simulator, are constant within every one of the 446 flights of DS02 and 553 of DS01. The damage therefore does not respond to the operating point within a flight, and the sensor map is the only possible source of any dependence on the operating point that the test detects.

Each cell is estimated twice, from its even and from its odd points. The mean of the two estimates is the signal; their half-difference measures the noise, and applying the same decomposition to it gives each effect its own noise floor. The decomposition is the ordinary two-way one over the cells, into a grand mean, a stage effect, an operating-point effect and an interaction, with sums of squares taken over cells and sensors and pooled over units. With 6 clusters and 2 stages the operating-point effect and the interaction each have 5 degrees of freedom per sensor and the stage effect 1, so the reported ratio of interaction to operating-point effect compares quantities with equal degrees of freedom. That ratio is 1.65 for DS02 and 1.84 for DS01. Against its own noise floor the interaction is 566 times larger on DS02 and 110 times on DS01, so it cannot be attributed to noise. The test was repeated at 3, 4, 6, 8 and 10 clusters with 2 stages, at 6 clusters with 3 stages and at 4 clusters with 4 stages, with at least 300 points per cell; the interaction exceeds the operating-point effect in every configuration, by a factor of at least 1.42.

S2 Information loss over a record and the best fixed indicator

S2.1 Loss over a complete record

Let the record consist of measurements (4) at instants $t_1, \dots, t_N$, with the noise independent across channels and across instants, and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1 + G}, \quad S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

Then the information about R that the fixed indicator discards over the complete record is

$$\ell_{\text{rec}} = -\frac{1}{2} \log(1 - WS). \quad (\text{S1})$$

In whitened coordinates the linearised record is $\tilde{y}_k = \mathbf{d}_k R + \epsilon_k$ with $\epsilon_k \sim N(0, I_d)$ independent, so the Fisher information it carries about R is $\sum_k \mathbf{d}_k^\top \mathbf{d}_k = \sum_k \|\mathbf{d}_k\|^2$ and an estimator using the complete record obtains $\frac{1}{2} \log(1 + G)$ nats. The fixed indicator returns the scalars $z_k = \mathbf{v}^\top \tilde{y}_k = (\mathbf{v}^\top \mathbf{d}_k) R + \mathbf{v}^\top \epsilon_k$, whose noise terms $\mathbf{v}^\top \epsilon_k$ are standard normal and independent because $\|\mathbf{v}\| = 1$, so the compressed record carries Fisher information $\sum_k (\mathbf{v}^\top \mathbf{d}_k)^2$. Since $\mathbf{v}^\top \mathbf{d}_k = \|\mathbf{d}_k\| \cos \psi_k$,

$$\text{Var}[R] \sum_k (\mathbf{v}^\top \mathbf{d}_k)^2 = \text{Var}[R] \sum_k \|\mathbf{d}_k\|^2 \cos^2 \psi_k = G(1 - S),$$

and subtracting the two mutual informations gives $\frac{1}{2} \log(1 + G) - \frac{1}{2} \log(1 + G(1 - S)) = -\frac{1}{2} \log \frac{1+G-GS}{1+G} = -\frac{1}{2} \log(1 - WS)$.

Equation (S1) has the form of (7), with the instantaneous signal-to-noise ratio replaced by its sum over the record and $\sin^2 \psi$ by its mean weighted by the signal energy. The averaging, however, differs: the time-averaged loss (9) averages the loss, whereas (S1) averages $\sin^2 \psi$ inside it, and the two differ and select different indicators. Minimising (S1) over $\|\mathbf{v}\| = 1$ maximises $\mathbf{v}^\top \left(\sum_k \mathbf{d}_k \mathbf{d}_k^\top \right) \mathbf{v}$, so the best fixed indicator for the record is the leading eigenvector of the integrated Fisher matrix.

S2.2 The best fixed indicator as a barycentre

Write $p_i = \hat{\mathbf{d}}_i^2$ and $q_i = \mathbf{v}_i^2$ for the information shares. By Appendix A.2, the components of $\hat{\mathbf{d}}$ are $\hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$, so $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\hat{\mathbf{d}}}_i$ gives the replicator equation of evolutionary dynamics, $\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$ [4]. Moreover, $\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and q , ψ is half the Fisher–Rao distance between them [5], and with $D_B = -\log \sum_i \sqrt{p_i q_i}$ the loss reads $\ell = -\frac{1}{2} \log[(1 - w) + w e^{-2D_B}]$. Let q^* minimise the time-averaged loss. (i) As $\gamma \rightarrow \infty$, $\ell \rightarrow D_B$ and q^* tends to the Bhattacharyya barycentre of the share path. (ii) As the path shortens, $\ell = \frac{w}{2} \psi^2 + O(\psi^4)$ and q^* tends to the w -weighted Fisher–Rao barycentre.

For (i), as $\gamma \rightarrow \infty$ we have $w \rightarrow 1$, and (7) gives

$$\ell \rightarrow -\frac{1}{2} \log \cos^2 \psi = -\log \sum_i \sqrt{p_i q_i} = D_B.$$

The convergence is uniform, and it is this uniformity that carries over to the minimisers. Positivity keeps p_τ interior, and $\sqrt{q_i} \geq q_i$ with $\sum_i q_i = 1$ gives $\cos \psi = \sum_i \sqrt{p_i q_i} \geq \min_i \sqrt{p_i}$ for every q in the simplex, so $\cos \psi$ is bounded away from zero uniformly in q and in τ and D_B remains finite. Both objectives are continuous on a compact set and uniformly close on it, so their minimisers converge. For (ii), $\bar{\ell} = \frac{1}{2} \bar{w} \mathbb{E}_\mu[\psi^2] + O(\psi^4)$ with $d\mu = w d\tau / \bar{w}$; minimising a μ -weighted mean square of a geodesic distance is the definition of the weighted Fréchet mean, and $\psi = \frac{1}{2} d_{\text{FR}}$ converts $\frac{1}{2} \bar{w} \mathbb{E}_\mu[\psi^2]$ into $\frac{1}{8} \bar{w} \mathbb{E}_\mu[d_{\text{FR}}^2]$.

Either barycentre of a nominal degradation path therefore provides a suitable initial indicator, and both lie near the midpoint of the arc used in the bound (12).

S3 Numerical verification

S3.1 Verification of the identities

The analytical results of the main text were verified numerically before they were used. The rotation identity (11) holds to relative error below 1.1×10^{-7} over the five input–channel pairs $(d, m) = (2, 1), (3, 2), (3, 1), (4, 2)$ and $(5, 3)$; the replicator form to 6.9×10^{-8} ; the rate of

change of the misalignment to 5.1×10^{-6} , limited by finite differencing; and the Bhattacharyya form of the loss to 1.1×10^{-16} .

S3.2 Non-generic systems that attain zero loss with a single input

Consider exponential shapes $h_i(x) = e^{\nu_i(x_i - x_i^0)}$ and Arrhenius factors written in reciprocal temperature, $\varphi_i(u) = e^{\theta_i - E_i u}$ with $u = 1/(kT)$. Under the ramp $u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 = \varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = r E_i / \varphi_i^0$ makes that velocity parallel to E , which is the direction of $\mathcal{B}$ (Section S4.1), and $b(u_t)$ recedes along $\mathcal{B}$ at the same rate. The two effects cancel, and the informative direction remains fixed with a single input, for any number of mechanisms.

Integrating this construction for $d = 2$ to 8 with one input holds the misalignment below 1.5×10^{-6} degrees and the mean loss below 10^{-15} , against 1.2×10^{-5} to 5.4×10^{-4} for the best constant input on the same system. These families have no prior variance of their own, so their losses are evaluated at a fixed $w = 0.9$ rather than at a value of γ supplied by the system; the bound has a margin of fifteen orders of magnitude and does not depend on this choice.

The family is not robust. Perturbing the shape coefficients ν by 1% raises the mean loss to 3.2×10^{-4} under the myopic policy, against 1.8×10^{-4} for the best constant input on the same perturbed system. With an error of one percent in the shape coefficients, steering therefore already performs worse than constant operation. Closing the loop halves this deterioration but does not remove it, so the fragility is inherent in the construction rather than in the open-loop schedule. Errors in the activation energies, by contrast, are tolerated up to about two percent.

S4 Conditions for zero loss and further results on the synthetic system

S4.1 Conditions for zero loss

Zero loss constrains the trajectory severely. If $\hat{\mathbf{d}}$ is held equal to $\mathbf{v}$, then $\dot{x} = \Sigma^{1/2} \mathbf{d} = \|\mathbf{d}\| \Sigma^{1/2} \mathbf{v}$, so the state can advance only along the straight segment

$$\mathcal{S} = \{x_0 + s \Sigma^{1/2} \mathbf{v} : 0 \leq s \leq s_f\} \quad (\text{S2})$$

from the initial state to the condemnation limit, and the operating point determines only the speed of progression. Zero loss is therefore attainable if and only if every state on $\mathcal{S}$ admits an admissible input that aligns the indicator, that is, $\mathbf{v} \in \mathcal{D}(x)$ for all $x \in \mathcal{S}$, with $\mathcal{D}(x)$ the set (13). The condition must be verified on $\mathcal{S}$ itself: an indicator may be reachable at every state of some other trajectory and nevertheless be unattainable, because the input that aligns it generates a different trajectory (Section S4.4 gives a two-state example).

Three consequences follow.

- Alignment at a given state imposes $d - 1$ scalar equations on m unknowns. With fewer than $d - 1$ independent operating variables, $\mathcal{D}(x_0)$ is a null set, and almost every indicator is already misaligned at the initial state.
- With $m \geq d - 1$ the alignment equations can be solved locally, but the solution must also remain within $\mathcal{U}$ along the whole of $\mathcal{S}$. The counting condition is thus necessary for generic data but not sufficient.

- Maintaining the misalignment at a constant, non-zero level is a single scalar condition on $\dot{u}$ and requires only one operating variable.

Systems for which zero loss is attainable with fewer inputs exist for any number of mechanisms, but they are non-generic: an error of one percent in their shape coefficients makes steering worse than constant operation (Section S3).

When the stress factors are separable, $\log \mathbf{d} = a(x) + b(u)$ up to a common constant, with $a_i = \log(h_i/\sigma_i)$ and $b_i = \log \varphi_i$. Every reachable set is then the translate by $a(x)$ of a single fixed set $\mathcal{B} = b(\mathcal{U})$, and zero loss requires the image of $\mathcal{S}$ under a to lie within a single translate of $-\mathcal{B}$. The actuation authority $\mathcal{B}$ can then be computed offline from an accelerated-life model, without any degradation record; Section 6 of the main text shows that this form, interpreted through the channel pairing (4), fails on raw turbofan sensors.

Proof of the conditions. Let each h_i be locally Lipschitz. By (7), $\bar{\ell} = 0$ if and only if $\hat{\mathbf{d}}(x_t, u_t) = \mathbf{v}$ for almost every t . Then $\dot{x}_t = \|\mathbf{d}_t\| \Sigma^{1/2} \mathbf{v}$ almost everywhere, so $x_t = x_0 + s(t) \Sigma^{1/2} \mathbf{v}$ with $s(t) = \int_0^t \|\mathbf{d}_\tau\| d\tau$ continuous and strictly increasing: the trajectory runs along $\mathcal{S}$. Since $\mathbf{v} \in \mathcal{D}(x_t)$ for almost every t , it lies in $\mathcal{D}(x)$ for x in a dense subset of $\mathcal{S}$, and the set $\{x : \mathbf{v} \in \mathcal{D}(x)\}$ is closed because $\mathcal{U}$ is compact and $\hat{\mathbf{d}}$ continuous; it therefore contains all of $\mathcal{S}$. Conversely, if $\mathbf{v} \in \mathcal{D}(x)$ on all of $\mathcal{S}$, the map $s \mapsto \{u \in \mathcal{U} : \hat{\mathbf{d}}(x_0 + s \Sigma^{1/2} \mathbf{v}, u) = \mathbf{v}\}$ has non-empty compact values and a closed graph, hence a measurable selection $\bar{u}(s)$ [6, Thm. 8.1.3]. As $\|\mathbf{d}\|$ is continuous and positive on the compact set $\mathcal{S} \times \mathcal{U}$, the time change $t(s) = \int_0^s \|\mathbf{d}(x_0 + \sigma \Sigma^{1/2} \mathbf{v}, \bar{u}(\sigma))\|^{-1} d\sigma$ is finite and strictly increasing, and with $s(t)$ its inverse the input $u(t) = \bar{u}(s(t))$ drives (1) along $\mathcal{S}$ with $\psi = 0$ up to the condemnation limit; local Lipschitz continuity of h ensures that this is the trajectory the input generates. For $m < d - 1$, $\mathcal{D}(x_0)$ is the image of the m -dimensional compact set $\mathcal{U}$ under a continuously differentiable map into the $(d - 1)$ -dimensional sphere, hence a null set that almost every $\mathbf{v}$ misses. In the separable case, $\mathbf{v} \in \mathcal{D}(x)$ reads $\kappa - a(x) \in \mathcal{B}$ with κ the class of $\log \mathbf{v}$ modulo the all-ones vector, and the condition on $\mathcal{S}$ is $a(\mathcal{S}) \subseteq \kappa - \mathcal{B}$.

S4.2 Effect of the time-shift assumption

Equation (3) is exact if and only if the population is spread along a single path. For the system of Section 5.2, with units instead generated by dispersing the degradation coefficients, the direction returned by the population regression of x on R lies about 45° from $-\varphi(u)h(x)$, across two families of dispersed coefficients and every spread examined; a control whose units are started on the reference trajectory, where (3) holds by construction, returns the same regression to within 0.2° . The misalignments considered in this paper amount to a few degrees. Extending the construction to populations dispersed in θ , for which the observations carry scatter that no value of R explains, remains an open problem.

S4.3 Loss at equal Fisher information of the record

Equation (10) can be illustrated on this system, at the steering indicator q^* of Section 5.2. For a constant input, a common gain on $\mathbf{d}$ changes $\|\mathbf{d}\|$ without rotating $\hat{\mathbf{d}}(\tau)$, so the score of the record can be equalised across inputs while the misalignment remains determined by the input. Let $\delta(u)$ denote the gain that equalises the score, defined by $\delta(u)^2 \|\mathbf{d}\|^2(u) = F$ with F its value at $u = 1.5$. It is realised by rescaling the duration for which the component operates under load, which is why it does not rotate $\hat{\mathbf{d}}$. By construction $\delta = 1$ at $u = 1.5$, and $\delta > 1$ wherever the raw score is smaller, so it is a multiplier on exposure rather than a fraction of it. Table S2 holds the score of the record equal to seven digits by tuning the exposure factor δ . The loss still spans

Table S2: Loss of the fixed indicator q^* at a record Fisher score held equal. The exposure factor δ , defined in the text, holds $\delta^2 \|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected; $\bar{I}_z = \delta^2 (\mathbf{v}^\top \mathbf{d})^2$ is the indicator’s own information, and rms β the root-mean-square misalignment. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\delta^2 \ \mathbf{d}\ ^2$	$\bar{I}_z$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	0.4055	11.80	0.0215
1.0	0.628	0.4307565	0.4207	8.98	0.0123
1.5	1.000	0.4307565	0.4170	11.61	0.0208
2.0	1.564	0.4307565	0.4050	15.60	0.0379
3.0	3.769	0.4307565	0.3421	26.98	0.1161

a factor 9.4. The information of the indicator itself, averaged in the same way, orders the five inputs exactly as the loss does, as (10) predicts.

The two objectives also favour different inputs: the raw score, before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and the loss is 1.7 times its value at $u = 1$. The extent of this disagreement is specific to the instance; the structural result is only that the information in the record does not determine the loss.

S4.4 An indicator reachable along a trajectory but unattainable on the segment

Let $d = 2$, $m = 1$, unit noise, $\varphi_i(u) = e^{\theta_i - E_i u}$ with $\theta = (0, 0.5)$, $E = (1, 2)$ and $\mathcal{U} = [0, 1.3]$, shapes $h_1 = 1$ and $h_2(x) = e^x$, $x_0 = 0$, $x_f = 0.9$, and the indicator with $\log(\mathbf{v}_2/\mathbf{v}_1) = -0.2$. It is aligned at state x by the input $u = 0.7 + x_2$, which is admissible exactly when $x_2 \leq 0.6$. Under the constant input $u = 1.3$ the system fails with $x_2 = 0.52$, so the indicator is reachable at every state that this trajectory visits. Zero loss is nevertheless unattainable: along the segment (S2), $x_2 = e^{-0.2} x_1$, and x_2 reaches 0.6 at $x_1 = 0.73$, before failure. The condition of Section S4.1 must therefore be verified on the segment rather than on the trajectory generated by an arbitrary fixed input.

S4.5 Attainability of zero loss with one and two inputs

With a single input, no direction lies in every reachable set of the test system. Along the closed loop at the steering indicator q^* the floor is largest at the start of life, where $\beta_{\min} = 4.52^\circ$; it decreases through mid-life to below the 0.01° resolution of the sampling, where the indicator is momentarily reachable, and increases again towards failure (Fig. S1(a)). Zero loss with q^* would require q^* to be reachable at every point of its segment $\mathcal{S}$, including x_0 , where the floor is largest; it is therefore unattainable under any input history. Fig. S2(a) illustrates the geometry.

A second independent input is introduced by setting $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m = 2 = d - 1$. Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction amounts to two linear equations in two unknowns, with a coefficient matrix of determinant 0.27 here, and hence a unique solution at every state. The target is taken as the direction occupied by the system at mid-life under the central operating point $(1.5, 1.5)$ of the box. Zero loss with this indicator confines the state to $\mathcal{S}$, and at every one of 600 points of that segment the aligning input lies inside the admissible box $[0, 3]^2$, at least 0.52 from its boundary. Along the segment it sweeps u_1 from 0.55 to 2.48 while u_2 stays between 1.32 and 1.70 (Fig. S1(b)). Under this feedback the state remains on $\mathcal{S}$ to machine precision and

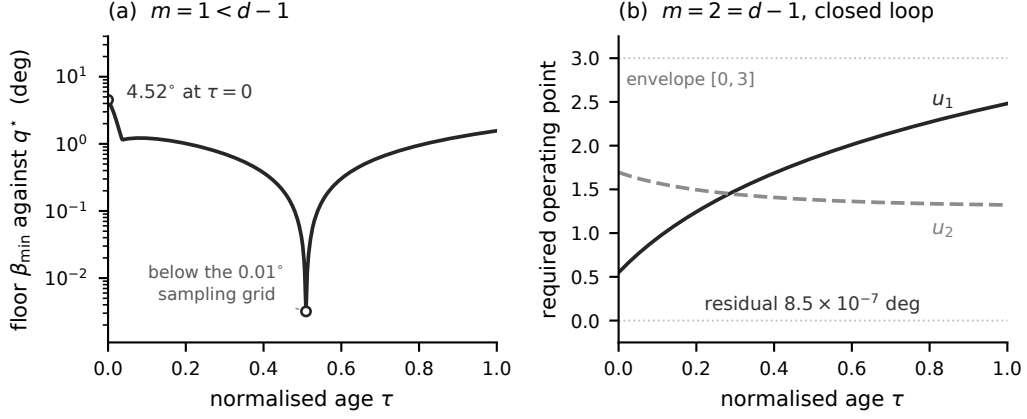


Figure S1: (a) With one input, the floor relative to q^* is largest at the start of life and falls below the sampling resolution near mid-life. (b) With two inputs, the mid-life direction is maintained exactly along the steered trajectory, which follows the segment $\mathcal{S}$; the required operating point, plotted against the normalised age of the steered trajectory, remains within $\mathcal{U}$.

the misalignment below 10^{-6} degrees, that is, zero to numerical precision, until failure. An independent 90×90 grid search over the box along the segment returns a floor of at most 0.152° , which reflects the resolution of the grid rather than a property of the system. For this system, therefore, the addition of one operating variable reduces the peak floor from 4.52° to zero, and zero loss is attained along the entire closed loop.

S4.6 Sensitivity to the condemnation limit

Raising x_f to unity leaves every qualitative conclusion unchanged and alters the constants as follows: the peak floor of Section S4.5 rises from 4.52° to 5.41° , and the factor of Section 5.2 at the steering indicator falls from 68 to 58. The constants reported in the main text therefore apply only at the stated condemnation limit.

S4.7 Computed fractional optimum

Dinkelbach's transform [7] converts the fractional problem (14) into a sequence of exit-time problems whose value functions solve a stationary Hamilton–Jacobi–Bellman equation on the state alone; these are discretised by a monotone semi-Lagrangian scheme [8]. The discrete Dinkelbach problem at q^* is solved on state grids of 25 to 137 points per axis, with the control grid of the policy. The scheme uses a fixed spatial step rather than a fixed time step, because degradation speeds vary by orders of magnitude across $\mathcal{U}$; since the drift is positive, the interpolation cell of every foot point has all its corners other than the node itself at a larger index sum, so the discrete equation is solved exactly by one sweep in decreasing index sum, and the optimal ratio by Dinkelbach's own iteration. The two steering factors of Section 5.2 are stable under refinement of the control grid: with control grids of 81 points the two factors become 72 and 50, and on 161 points 72 and 50 again. The values converge at first order in the mesh width: fitted to the four finest grids, the order is 0.99. Extrapolated from those four, $\lambda_\infty^* = 1.65 \times 10^{-4}$. The myopic policy attains 1.68×10^{-4} and is realisable, so it bounds the optimum from above, and the optimum lies between the two values. The myopic policy therefore sits 1.4% above the optimum as extrapolated. Fitting the four next-finest grids instead, or repeating the computation on control grids of 81 and 161 points, puts the gap between 1.4% and 2.0%, while refining the

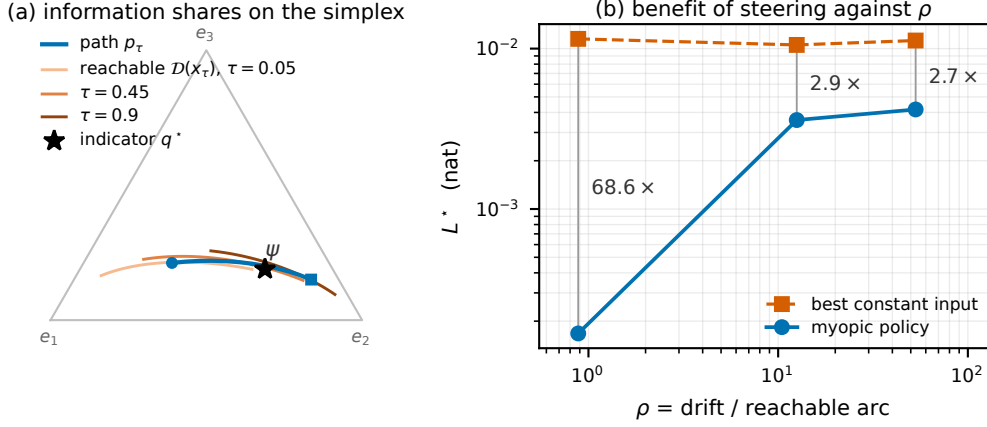


Figure S2: (a) The information-share path p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the steering indicator q^* , which lies off the arcs near the start of life and near failure. (b) Reduction of the information loss achieved by steering, relative to the best constant input at the same indicator, as a function of ρ , the drift of the reachable arc relative to its length.

control grid from 41 to 81 points lowers both the myopic value and the extrapolated optimum by more than 4%.

For this system, therefore, the fractional form of (14) affects the formulation of the problem and its certification, but not measurably its solution: a policy that considers only the numerator is optimal to within the resolution of the computation. The lower end of the interval rests on the extrapolation and is not a bound, since even the finest grid returns a value 27% above it. The computed value is subject to three checks: the Dinkelbach function must vanish at the computed λ^* , λ^* must not exceed the value of a realisable policy, and the grid sequence must extrapolate consistently.

S4.8 Screening by actuation authority and drift

Whether steering can reduce the loss substantially can be assessed from the system model before any indicator is designed. Across 299 random systems of the separable model, the reachable-set width ranks the systems according to whether steering cuts the loss fivefold with an area under the receiver-operating curve of 0.73; the drift of the informative direction contributes additional information only through its interaction with the width, as the rest of this section shows.

The reachable set $\mathcal{D}(x)$ is an arc, and it suggests two natural indicators of whether steering is worthwhile: its mean length, which is the actuation authority available to follow the informative direction, and the drift of its midpoint over the life, which measures how far that direction moves. Let ρ denote the drift divided by the length, which weighs the two against each other.

Table S3 varies the spread of the activation energies, which by (11) determines that authority. As the arc shrinks, the attainable loss L^* rises by a factor 25 and the value of control falls from 68 to 2.7 against the best constant input at the same indicator, and from 47 to 1.9 against a constant design with its own: with weak authority, steering has little effect. Fig. S2(b) plots the two against ρ . In this family the drift barely changes, staying between 24° and 30° while the arc spans nearly two decades, so ρ is effectively the reciprocal of the arc and the table cannot distinguish which of the two matters. Both can be computed from a nominal degradation model before any indicator is designed.

Table S3: Authority and drift in a one-parameter family. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under the myopic policy at the indicator optimised for it. The first gain is its ratio to the best constant input at that indicator, the second to the best constant input with the indicator optimised for it.

E	arc	ρ	L^*	gain	own-indicator gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.7×10^{-4}	68×	47×
(0.80, 0.82, 0.79)	1.93°	12.5	3.6×10^{-3}	2.9×	2.2×
(0.80, 0.805, 0.798)	0.45°	53.1	4.2×10^{-3}	2.7×	1.9×

Three systems that differ only in their activation energies do not establish a screening rule, so the index was tested on 299 further systems drawn at random: two to five mechanisms, one or two independently adjustable inputs, and the pre-exponentials, activation energies, shape coefficients, input envelope, initial state and signal-to-noise ratio varied together. The gain is measured as in the first gain column of Table S3, against the best constant input at the steering indicator, which on the three systems of that table overstates it by factors of 1.3 to 1.4 relative to a constant design with its own indicator. The rank correlation between ρ and the gain is -0.36 , with a bootstrap 95% interval from -0.46 to -0.24 , and it keeps its sign inside every stratum of the mechanism count and of the input count, so it is not an artefact of either.

Used as a classifier of whether steering reduces the loss fivefold, an outcome observed for 32% of the systems, ρ has an area under the receiver-operating curve of 0.73, with interval 0.67 to 0.79. The reachable arc alone scores 0.73 as well, and the drift alone 0.53, with interval 0.46 to 0.60, which includes the value 0.5 of a random classifier.

Equal areas do not imply equal merit, and the question is resolved by a model that contains both as special cases: a logistic regression on the logarithms of arc and drift as separate terms, of which the arc alone and the ratio ρ are each a constrained case. The drift adds nothing to the arc as a main effect ($p = 0.47$), and constraining the two to enter as a ratio fits significantly worse than the arc alone ($p = 7.7 \times 10^{-5}$; Akaike information 351 against 336). Both results hold in each input count separately. The drift does carry information, but through the arc ($p = 3.0 \times 10^{-4}$ for their interaction), and in opposite directions at the two ends. Among systems of low authority the share repaying fivefold falls from 0.18 to 0.08 as the drift rises from its lowest to its highest third; among systems of high authority it rises from 0.48 to 0.61. Drift that exceeds the authority defeats steering, whereas drift that the authority can follow is what makes steering useful. The ratio captures only the first of these effects. This also explains why the lowest tenth of ρ repays fivefold only 43% of the time: there the drift is small, a median of 2.7° against 14° over the sample.

The ranking is confirmed out of sample. Over repeated ten-fold cross-validation the arc and ρ each rank unseen systems with an area of 0.72, and a model adding the drift and its interaction with the arc reaches 0.74, higher in every repeat. The appropriate screening quantity is therefore the arc, that is, the authority itself, refined by the drift only through their interaction; ρ remains a convenient description of the family of Table S3, in which the drift is nearly fixed. The first row of that table is not typical of its own regime: among sampled systems within a factor 1.4 of its ρ , the median gain is 5.2 rather than 68.

S4.9 Effect of the loss on a downstream estimator

Equation (8) expresses the loss in terms of the error of the estimate prescribed by the model itself. Since a practitioner does not use that estimate, the ordering is examined against a pro-

cedure that uses none of the construction: the indicator is sampled over the last 40% of life, the observed history is matched against the indicator’s own reference curve, and the remaining life is read from the best-fitting time offset. This estimator uses only the scalar signal; it has no access to $\hat{\mathbf{d}}$, γ or the loss. It is the estimator of Table 2.

On one constant-input path of the synthetic system, over twelve indicators spanning $\bar{\ell}$ from 0.012 to 0.37, its median absolute error, one of the standard prognostic accuracy measures [9], increases with the loss, at Spearman correlation 0.89, and the best-aligned indicator beats the worst by a factor 1.5. Since twelve indicators constitute a small sample, both quantities are reported with intervals obtained by resampling the 160 units behind each median: 0.34 to 0.90 for the correlation, and 1.40 to 2.03 for the factor. The ordering is statistically significant (a one-sided permutation test gives $p = 1.5 \times 10^{-4}$), but its strength is uncertain, and the point estimate lies near the upper end of its own interval. The factor is far smaller than the 31-fold spread in $\bar{\ell}$, as expected: the loss is a logarithmic information measure, whereas the error is a time, and an estimator using the whole history recovers much of what any single instant discards.

The same protocol on 120 further systems, drawn at random as in Section S4.8, gives correlations between 0.87 and 0.99, with the system above among the weaker. That design, however, tilts every indicator towards the same direction, so that loss and error both increase with a single parameter and their agreement is largely structural. Tilting instead by a random amount of up to 60° in a random direction weakens the median correlation to 0.74, with the interval excluding zero on 84% of systems. It also reveals a limitation of the experiment. On these systems the angle between an indicator and the mid-life informative direction ranks the indicators identically to the loss on 82% of systems, with neither consistently better on the remainder, because the direction rotates little compared with the differences between the indicators. The experiment is therefore evidence that the ordering of the loss carries over to an estimator built from none of this construction; it is not evidence that the time-averaged loss improves on a mid-life snapshot of the misalignment, which would require a system whose direction rotates by more than its candidate indicators differ.

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